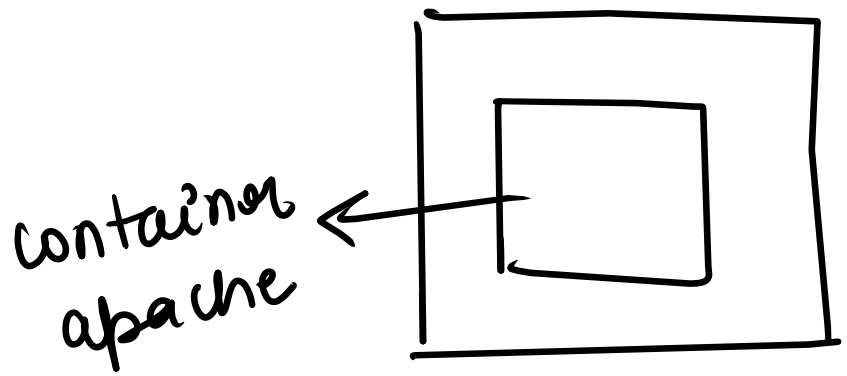
We need to create a pod with apache (web-server) as a container.



pod

pod are only created in the nodes (never in master)

We need to write a script which we call it as a manifest.

Manifest is written in master machine & pods are created in the nodes.

manifest is written in yaml (yet another markup language

new full form:- yaml aint markup language)

very simple engish language

```
- Institute:
    - Devopsinstitute:
        - Staragile
            - Trainers:
                Akshat
                John
                Joseph
    - MBAinstitute:
        - Times:
            - Trainers:
                Aman
    - JEEinstitute:
        - Allen:
            - Trainers:
                RKSIr
                MANISH SIR
```

Indentation:- one below another

```
root@ip-172-31-6-141:/home/ubuntu# kubectl get nodes
```

kubernetes controller

get the nodes in the cluster.

```
root@ip-172-31-6-141:/home/ubuntu# kubectl get nodes
NAME                STATUS   ROLES           AGE   VERSION
ip-172-31-15-78     Ready    <none>          18m   v1.29.15
ip-172-31-2-109     Ready    <none>          18m   v1.29.15
ip-172-31-6-141     Ready    control-plane   24m   v1.29.15
root@ip-172-31-6-141:/home/ubuntu#
```

lets create a pod with apache container in it

```
# vi pod1.yml
```

manifest file name
It could be any file
name ending with .yml

we are creating the pod

Apiversion depends on the Kind. you cannot change apiversion

https://blog.nashtechglobal.com/what-is-apiversion-in-kubernetes-part-1/

```yaml
kind: Pod
apiVersion: v1
metadata:
  name: myfirstpod
spec:
  containers:
    - name: apachecon
      image: httpd
      ports:
        - containerPort: 80
```

The name of the pod is myfirstpod

The pod will have containers

The name of the pod is apachecon

The image is httpd of container

what is the specification of pod

App is running on port 80 of the container

```yaml
kind: Pod
apiVersion: v1
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  name: myfirstpod
spec:
  containers:
    - name: apachecon
      image: httpd
      ports:
        - containerPort: 80
```

we are creating a pod

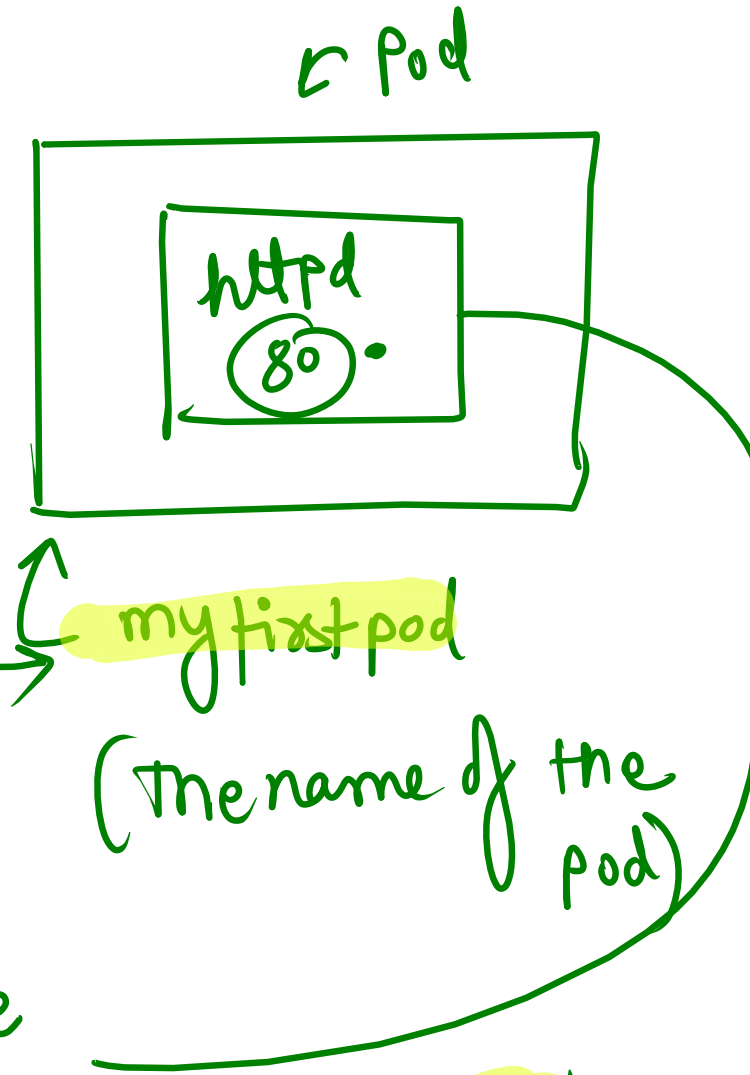
fixed and depends on kind

(Base Image)

The default app is working on port 80

Container is there Inside pod

The name of container is apachecon



```yaml
kind: Pod
apiVersion: v1
metadata:
  name: myfirstpod
spec:
  containers:
    - name: apachecon
      image: httpd
      ports:
        - containerPort: 80
```

we have containers Inside the pod

httpd is apache Image

apachecon

What is there Inside the pod

pod

myfirstpod

name

```
root@ip-172-31-6-141:/home/ubuntu# kubectl apply -f pod1.yml
```

To execute

file

filename

you will see list of pods

```
root@ip-172-31-6-141:/home/ubuntu# kubectl get pods -o wide
NAME          READY   STATUS    RESTARTS   AGE    IP               NODE
myfirstpod    1/1     Running   0          118s   192.168.9.129    ip-172-31-2-109
root@ip-172-31-6-141:/home/ubuntu#
```

pvt ip of pod

In node 2 the pod is created